SOFTWARE

Complete MCQ Question Bank

160 Questions · 3 Difficulty Levels · 9 Question Types

For Competitive & Teacher Recruitment Exams

OAVS · KVS · NVS · DSSSB · OSSTET · CTET · TGT/PGT Computer Science

160	53	55	52
Total Questions	Basic Level	Intermediate	Advanced

Topics Covered: Introduction to Programming · Flowcharts & Algorithms · System Software · Application Software · Firmware · Assembly Language · High-Level Languages · Stored Program Concept · MS Office Packages · Language Translators · Software Types & Licensing · SDLC & Testing

Quick Reference — Flowchart Symbols

Symbol	Shape	Purpose
Terminal	Oval / Ellipse	Start or Stop
Process	Rectangle	Operation or computation
Decision	Diamond (Rhombus)	Yes/No or True/False condition
Input/Output	Parallelogram	Read input or display output
Connector	Small Circle	Connect flow on same/different page
Flow Line	Arrow	Direction of process flow

Quick Reference — Language Generations

Generation	Name	Example	Translator
1GL	Machine Language	Binary (0s and 1s)	None
2GL	Assembly Language	MOV, ADD, SUB	Assembler
3GL	High-Level Language	C, Java, FORTRAN, COBOL	Compiler / Interpreter
4GL	Very High-Level / Query	SQL, MATLAB, R	Interpreter
5GL	AI / Natural Language	Prolog, LISP	Inference Engine

Basic Level — 53 Questions

Q1 Definition — Intro to Programming

What is a computer program?

A. A set of instructions for hardware

B. A set of instructions that directs a computer to perform specific tasks

C. A type of hardware component

D. An operating system feature

Ans: B A computer program is a set of coded instructions that directs a computer to perform specific operations or solve a problem.

Q2 Definition — Flowchart

A flowchart is a:

A. Programming language

B. Diagrammatic representation of an algorithm

C. Type of database

D. Hardware circuit diagram

Ans: B A flowchart is a graphical/pictorial representation of an algorithm using standard symbols.

Q3 Concept — Algorithm

Which of the following is NOT a property of a good algorithm?

A. Finiteness

B. Definiteness

C. Infinity

D. Effectiveness

Ans: C A good algorithm must be finite, definite, effective, and have proper input/output. Infinity is NOT a property.

Q4 Concept — System Software

Which of the following is system software?

A. MS Word

B. Operating System

C. Tally

D. VLC Player

Ans: B System software includes OS, device drivers, utilities. MS Word, VLC, Tally are application software.

Q5 Definition — System Software

System software is defined as:

A. Software used for business applications

B. Software that manages hardware and provides platform for application software

C. Software used for gaming

D. Software that connects the internet

Ans: B System software manages and controls hardware resources and provides a platform for running application software.

Q6 Definition — Application Software

Application software is designed to:

A. Manage computer hardware

B. Perform specific tasks for the end user

C. Control the operating system

D. Compile programs

Ans: B Application software performs specific tasks directly for end users — word processing, spreadsheets, browsing, etc.

Q7 Concept — Firmware

Firmware is stored in:

A. RAM

B. Hard Disk

C. ROM/Flash memory

D. Cache memory

Ans: C Firmware is stored in non-volatile memory (ROM or Flash). It provides low-level control for hardware devices.

Q8 | Concept | Assembly Language

Assembly language uses _____ to represent machine instructions.

A. Binary digits

B. Hexadecimal only

C. Mnemonics

D. English sentences

Ans: C Assembly language uses mnemonics (symbolic codes like MOV, ADD, SUB) instead of binary 0s and 1s.

Q9 | Concept | High-Level Language

High-level languages are closer to:

A. Machine language

B. Assembly language

C. Human language

D. Firmware

Ans: C High-level languages (Python, Java, C++) use English-like syntax and are closer to human language.

Q10 | Concept | Stored Program Concept

The stored program concept was proposed by:

A. Charles Babbage

B. Alan Turing

C. John von Neumann

D. Dennis Ritchie

Ans: C John von Neumann proposed the stored program concept — program instructions and data stored in the same memory.

Q11 | Fill-Blank | MS Office

The default file extension for MS Word 2016 and later is _____.

A. .doc

B. .docx

C. .txt

D. .odt

Ans: B MS Word 2007 and later uses .docx (Office Open XML format). The older .doc was used in Word 97-2003.

Q12 Fill-Blank MS Office

MS Excel is used primarily for _____.

A. Word processing

B. Presentations

C. Spreadsheet and data analysis

D. Database management

Ans: C MS Excel is a spreadsheet application used for data entry, calculations, charts, and data analysis.

Q13 Concept Flowchart

Which symbol is used for a decision in a flowchart?

A. Rectangle

B. Oval/Ellipse

C. Diamond

D. Parallelogram

Ans: C The diamond (rhombus) shape represents a decision (Yes/No or True/False) in a flowchart.

Q14 Concept Flowchart

The oval/ellipse symbol in a flowchart represents:

A. Process

B. Decision

C. Input/Output

D. Start/Stop (Terminal)

Ans: D The oval or ellipse symbol represents the terminal (Start or Stop) in a flowchart.

Q15 Concept Flowchart

The parallelogram symbol in a flowchart represents:

A. Process

B. Input/Output

C. Decision

D. Connector

Ans: B The parallelogram represents Input or Output operations in a flowchart.

Q16 Concept — Algorithm

An algorithm is:

A. A programming language

B. A step-by-step procedure to solve a problem

C. A flowchart only

D. A binary code

Ans: B An algorithm is a finite, well-defined, step-by-step procedure to solve a particular problem.

Q17 Concept — Machine Language

Machine language uses:

A. Mnemonics

B. English words

C. Binary digits (0s and 1s)

D. Symbols only

Ans: C Machine language (1st generation) uses binary digits (0s and 1s) and is directly understood by the CPU.

Q18 Concept — MS Office

Which MS Office application is used for presentations?

A. MS Word

B. MS Excel

C. MS Access

D. MS PowerPoint

Ans: D MS PowerPoint is the presentation software in the MS Office suite.

Q19 Concept — MS Office

MS Access is a:

A. Spreadsheet application

B. Word processor

C. Database management system

D. Presentation tool

Ans: C MS Access is a relational database management system (RDBMS) included in MS Office.

Q20 Concept — Firmware

BIOS is an example of:

A. Application software

B. System software

C. Firmware

D. Freeware

Ans: C BIOS (Basic Input/Output System) is firmware stored on a chip on the motherboard.

Q21 Concept — Algorithm

Pseudo-code is:

A. A programming language like C

B. An informal high-level description of algorithm logic

C. Binary code

D. A type of flowchart

Ans: B Pseudo-code is an informal, human-readable description of an algorithm using plain English constructs.

Q22 Concept — MS Office

Which of the following is NOT part of the MS Office suite?

A. MS Word

B. MS Excel

C. Adobe Photoshop

D. MS PowerPoint

Ans: C Adobe Photoshop is a product of Adobe Inc., not Microsoft.

Q23 Fill-Blank — Algorithm

The _____ of an algorithm refers to its ability to produce the correct output.

A. Efficiency

B. Correctness

C. Finiteness

D. Input

Ans: B Correctness means the algorithm must produce the right output for all valid inputs.

Q24 Concept — Machine Language

Machine language belongs to which generation of programming languages?

A. 1GL

B. 2GL

C. 3GL

D. 4GL

Ans: A Machine language (binary: 0s and 1s) is the first generation language (1GL).

Q25 Concept — Assembly Language

Assembly language belongs to which generation?

A. 1GL

B. 2GL

C. 3GL

D. 5GL

Ans: B Assembly language (using mnemonics) is the second generation language (2GL).

Q26 Concept — MS Office

The extension of an MS PowerPoint file (2007 and later) is:

A. .ppt

B. .pptx

C. .ppsx

D. .pot

Ans: B .pptx is the default extension for PowerPoint presentations in Office 2007 and later.

Q27 Concept — MS Office

MS Outlook is primarily used for:

A. Spreadsheets

B. Email and calendar management

C. Presentations

D. Database management

Ans: B MS Outlook handles email, calendar events, contacts, and tasks.

Q28 Fill-Blank MS Office

In MS Excel, the formula to add a range of cells A1 to A10 is _____.

A. =ADD(A1:A10)

B. =SUM(A1:A10)

C. =TOTAL(A1:A10)

D. =PLUS(A1:A10)

Ans: B =SUM(A1:A10) is the correct Excel formula to add all values in the range A1 to A10.

Q29 Concept High-Level Language

Which was the first high-level programming language?

A. COBOL

B. FORTRAN

C. BASIC

D. Pascal

Ans: B FORTRAN (FORmula TRANslation), developed by IBM in 1957, is the first high-level programming language.

Q30 Concept Software

What distinguishes hardware from software?

A. Hardware is intangible; software is tangible

B. Hardware is tangible physical components; software is intangible programs and data

C. They are identical

D. Hardware needs no electricity

Ans: B Hardware = physical tangible components (CPU, RAM). Software = intangible programs and instructions.

Q86 Concept Software

Which of the following is an example of utility software?

A. MS Word

B. Disk Defragmenter

C. Mozilla Firefox

D. Photoshop

Ans: B Utility software (Disk Defragmenter, Antivirus, File Manager) maintains and optimizes the system.

Q87 Concept — MS Office

The shortcut key to create a new slide in MS PowerPoint is:

A. Ctrl+N

B. Ctrl+M

C. Ctrl+S

D. Ctrl+T

Ans: B Ctrl+M inserts a new slide in MS PowerPoint. Ctrl+N creates a new presentation.

Q88 Concept — Flowchart

The connector symbol in a flowchart (small circle) is used to:

A. Represent a decision

B. Connect two parts of a flowchart on the same or different pages

C. Mark the start

D. Represent input

Ans: B The small circle (connector) links different parts of a flowchart, especially across pages.

Q89 Concept — Algorithm

What does 'debugging' mean?

A. Writing a program

B. Finding and fixing errors in a program

C. Compiling a program

D. Designing an algorithm

Ans: B Debugging is the process of finding, analyzing, and fixing errors (bugs) in a program.

Q90 Fill-Blank — MS Office

The rectangle symbol in a flowchart represents a _____.

A. Decision

B. Process/Operation

C. Start/Stop

D. Input/Output

Ans: B The rectangle represents a Process or Operation step (e.g., Compute, Set, Calculate).

Q91 Concept MS Office

Which shortcut key in MS Word is used for 'Undo'?

A. Ctrl+Y

B. Ctrl+Z

C. Ctrl+X

D. Ctrl+U

Ans: B Ctrl+Z is the Undo shortcut across all MS Office applications.

Q92 Concept MS Office

Which shortcut key is used for 'Redo' in MS Office?

A. Ctrl+Z

B. Ctrl+R

C. Ctrl+Y

D. Ctrl+D

Ans: C Ctrl+Y performs the Redo operation in MS Office applications.

Q93 Concept Software Types

Which generation of software uses natural language processing?

A. 1st generation

B. 2nd generation

C. 3rd generation

D. 5th generation

Ans: D 5th generation software uses AI and natural language processing, such as voice assistants and expert systems.

Q94 Concept Algorithm

The number of comparisons in binary search for n=16 elements (worst case) is:

A. 4

B. 5

C. 8

D. 16

Ans: B Binary search worst case = log2(n) comparisons. log2(16) = 4... actually log2(16)+1 = 5 comparisons. Correct: 5.

Q95 Concept MS Office

Which tab in MS Word contains the 'Page Margins' option?

A. Home

B. Insert

C. Layout/Page Layout

D. Review

Ans: C Page margins are set under the Layout (or Page Layout) tab in MS Word.

Q116 Concept Software Types

Which of the following is NOT a type of system software?

A. Operating System

B. Device Driver

C. Compiler

D. MS Excel

Ans: D MS Excel is application software. OS, device drivers, and compilers are all system software.

Q117 Concept Algorithm

Which symbol represents a 'connector on a different page' in a flowchart?

A. Circle

B. Pentagon

C. Rectangle

D. Diamond

Ans: B An off-page connector is represented by a pentagon (house shape) or a specific off-page connector symbol.

Q118 Concept MS Office

What is the maximum number of rows in MS Excel 2016?

A. 65,536

B. 1,048,576

C. 524,288

D. 2,097,152

Ans: B MS Excel 2007 and later supports 1,048,576 rows and 16,384 columns per worksheet.

Q131 Concept — MS Office

Which key is used to start a presentation (slideshow) from the beginning in MS PowerPoint?

A. F1

B. F5

C. F7

D. F10

Ans: B F5 starts the slideshow from the first slide. Shift+F5 starts from the current slide.

Q132 Definition — Software

Which of the following best defines 'software'?

A. Physical parts of a computer

B. A set of programs, data, and instructions that tell a computer what to do

C. The monitor and keyboard

D. The power supply unit

Ans: B Software is the collection of programs, data, and instructions that direct the hardware to perform tasks. It is intangible.

Q133 Concept — Algorithm

Which of the following is the correct sequence of steps in problem-solving using a computer?

A. Coding → Algorithm → Flowchart → Testing

B. Problem Definition → Algorithm → Flowchart → Coding → Testing

C. Flowchart → Problem → Coding → Algorithm

D. Testing → Coding → Algorithm → Flowchart

Ans: B The correct sequence is: Problem Definition → Algorithm → Flowchart → Coding → Testing → Documentation.

Q134 | Fill-Blank | MS Office

In MS Word, the shortcut key to insert a page break is _____.

A. Ctrl+Enter

B. Ctrl+P

C. Ctrl+B

D. Ctrl+N

Ans: A Ctrl+Enter inserts a manual page break in MS Word, forcing the next content to start on a new page.

Q135 | Concept | High-Level Language

Which of the following is a scripting language used for web development?

A. C

B. FORTRAN

C. PHP

D. COBOL

Ans: C PHP (Hypertext Preprocessor) is a popular server-side scripting language used for web development.

Q136 | Concept | System Software

Which of the following is an example of an open-source operating system?

A. Windows 11

B. macOS

C. Linux

D. MS-DOS

Ans: C Linux is a free, open-source operating system kernel. Its distributions (Ubuntu, Fedora) are widely used.

Q137 | Concept | Flowchart

In a flowchart, which symbol is used to represent a process that is defined elsewhere (a sub-process)?

A. Diamond

B. Rectangle with double vertical lines (Pre-defined process)

C. Oval

D. Parallelogram

Ans: B The pre-defined process symbol (rectangle with double vertical bars on sides) represents a named process defined elsewhere, such as a subroutine or function.

Q138 | Definition | Assembly Language

A mnemonic is:

A. A type of hardware register

B. A memory-aiding abbreviation used in assembly language (e.g., MOV, ADD)

C. A type of compiler

D. A machine language instruction

Ans: B A mnemonic is a short, human-readable abbreviation for a machine instruction used in assembly language, e.g., MOV for move, ADD for addition.

Q139 | Fill-Blank | MS Office

The default file extension for MS Excel 2007 and later is _____.

A. .xls

B. .xlsx

C. .csv

D. .xlsm

Ans: B .xlsx is the default Excel format from Office 2007 onwards (Open XML). .xls was the older Excel 97-2003 format.

Q140	Concept	Stored Program Concept

In a computer system, which component stores currently running programs and data?

A. Hard Disk Drive

B. ROM

C. RAM (Primary Memory)

D. Cache only

Ans: C RAM (Random Access Memory) is the primary/main memory that stores currently running programs and data. It is volatile — data is lost when power is off.

Intermediate Level — 55 Questions

Q31 Concept — Software Types

Which of the following best describes middleware?

A. Software between OS and hardware

B. Software that connects different applications or services

C. Software used in the middle of programming

D. A type of firmware

Ans: B Middleware bridges different applications/services, enabling communication and data management.

Q32 Concept — Language Translator

An assembler translates:

A. High-level language to machine code

B. Assembly language to machine code

C. Machine code to assembly

D. High-level to assembly

Ans: B An assembler translates assembly language (mnemonics) into machine language (binary code).

Q33 Concept — Language Translator

A compiler:

A. Translates one line at a time

B. Translates the entire program at once before execution

C. Is used only for assembly language

D. Is slower than interpreter for repeated execution

Ans: B A compiler translates the entire source code into object code at once. Execution is then fast.

Q34 Concept — Language Translator

An interpreter:

A. Translates the whole program before running

B. Translates and executes one line at a time

C. Generates object code

D. Is faster than a compiler for repeated execution

Ans: B An interpreter translates and executes the program line by line — slower for repeated runs but easier to debug.

Q35 | Statement | Stored Program Concept

Consider:
(I) Program and data are stored in the same memory.
(II) CPU fetches, decodes and executes instructions from memory.
Which is/are correct?

A. Only I

B. Only II

C. Both I and II

D. Neither I nor II

Ans: C Both are correct. The stored program concept (von Neumann) stores program+data in main memory; CPU fetch-decode-executes.

Q36 | Application | Algorithm

What is the time complexity of a linear search algorithm?

A. O(1)

B. O(log n)

C. O(n)

D. O(n^2)

Ans: C Linear search examines each element one by one: O(n) time complexity.

Q37 | Concept | Software Types

Which of the following is general-purpose application software?

A. Payroll software

B. Airline reservation system

C. MS Word

D. Inventory management system

Ans: C General-purpose software (MS Word, Excel) serves a wide range of users.

Q38 | Concept | High-Level Language

Which of the following is a 4th generation language (4GL)?

A. Assembly

B. FORTRAN

C. SQL

D. C++

Ans: C SQL is a 4GL — non-procedural, closer to natural language. FORTRAN is 3GL; Assembly is 2GL.

Q39 Application MS Office

Which shortcut key is used to save a file in MS Office?

A. Ctrl+P

B. Ctrl+S

C. Ctrl+X

D. Ctrl+Z

Ans: B Ctrl+S is the universal shortcut to save a file in all MS Office applications.

Q40 Application MS Office

The function VLOOKUP in MS Excel is used to:

A. Add values vertically

B. Look up a value in the leftmost column of a table

C. Format cells

D. Sort data

Ans: B VLOOKUP searches for a value in the first column of a range and returns a value in the same row.

Q41 Concept Flowchart

A system flowchart shows:

A. Detailed logic of a single program

B. Overall data flow through an entire system

C. Only input/output operations

D. Source code logic

Ans: B A system flowchart shows overall data flow through an entire system without implementation details.

Q42 | Assertion-Reason | Software

Assertion (A): Open-source software can be freely used, modified, and distributed.
Reason (R): Open-source software has no source code.
Choose the correct option:

A. Both A and R are true, R explains A

B. A is true but R is false

C. Both A and R are false

D. A is false but R is true

Ans: B Open-source CAN be freely used/modified/distributed (A true). But it DOES have source code — R is false.

Q43 | Concept | Software Types

Freeware is:

A. Free software with source code

B. Software available free of cost but without source code

C. Open-source software

D. Trial software

Ans: B Freeware is available free of cost, but source code is NOT provided. Examples: Skype, Adobe Reader.

Q44 | Match | Language Translator

Match the translator with the language:
P. Assembler Q. Compiler R. Interpreter
1. Python 2. Assembly 3. C program

A. P-2, Q-3, R-1

B. P-1, Q-2, R-3

C. P-3, Q-1, R-2

D. P-2, Q-1, R-3

Ans: A Assembler: Assembly(2); Compiler: C(3); Interpreter: Python(1).

Q45 Concept — Algorithm

Which data structure is used in Breadth First Search (BFS)?

A. Stack

B. Queue

C. Tree

D. Graph only

Ans: B BFS uses a Queue (FIFO) to explore nodes level by level. DFS uses a Stack.

Q46 Application — MS Office

In MS PowerPoint, the 'Slide Transition' feature controls:

A. Font style changes

B. Animation within a single slide

C. Movement from one slide to another

D. Adding background music

Ans: C Slide Transition controls how one slide changes to the next during a presentation.

Q47 Concept — Firmware

UEFI is the modern replacement for:

A. RAM

B. BIOS

C. ROM

D. Hard Disk

Ans: B UEFI (Unified Extensible Firmware Interface) replaces traditional BIOS. It supports larger drives and secure boot.

Q48 Concept — Software Types

Shareware is:

A. Completely free software

B. Software distributed free for trial, then requires purchase

C. Software with open source code

D. Malicious software

Ans: B Shareware is distributed free for a limited trial period, after which payment is required.

Q49 Application MS Office

Which function in Excel counts only non-empty cells?

A. COUNT

B. COUNTA

C. COUNTBLANK

D. COUNTIF

Ans: B COUNTA counts all non-empty cells. COUNT counts only numeric values.

Q50 Concept Language Translator

Which phase of a compiler checks for grammatical errors in source code?

A. Lexical analysis

B. Syntax analysis (Parsing)

C. Semantic analysis

D. Code generation

Ans: B Syntax analysis (parsing) checks if the source code conforms to grammar rules of the programming language.

Q51 Concept High-Level Language

COBOL was developed for:

A. Scientific computations

B. Business data processing

C. System programming

D. Web development

Ans: B COBOL (Common Business-Oriented Language) was developed in 1959 for business data processing.

Q52 Concept High-Level Language

Which of the following is an object-oriented programming language?

A. FORTRAN

B. COBOL

C. Java

D. Assembly

Ans: C Java is fully object-oriented. FORTRAN and COBOL are procedural; Assembly is low-level.

Q53 Concept — System Software

Which of the following is NOT a function of an operating system?

A. Memory management

B. File management

C. Word processing

D. Process management

Ans: C Word processing is application software. OS handles memory, files, processes, devices.

Q54 Application — MS Office

Macro in MS Office is used to:

A. Format text manually

B. Automate repetitive tasks using recorded/programmed sequences

C. Create tables

D. Print documents

Ans: B A Macro records/programs a sequence of commands to automate repetitive tasks.

Q55 Concept — Software Types

Which type of software license allows study, change, and distribution freely?

A. Proprietary license

B. Freeware license

C. Open-source license

D. Shareware license

Ans: C Open-source licenses (GPL, MIT, Apache) allow users to view, modify, and distribute source code freely.

Q56 Application — MS Office

Mail Merge in MS Word is used to:

A. Merge two Word documents

B. Create personalized letters/documents for multiple recipients

C. Format email attachments

D. Send emails directly

Ans: B Mail Merge combines a template with a data source to create personalized documents for multiple recipients.

Q57 Application MS Office

Which view in MS Word shows the document exactly as it will print?

A. Draft view

B. Web layout view

C. Print Layout view

D. Outline view

Ans: C Print Layout view shows the document as it will appear when printed, with margins and page breaks.

Q58 Concept Software Types

Malware is short for:

A. Machine language software

B. Malicious software

C. Multi-layered software

D. Management software

Ans: B Malware = Malicious Software. Examples: viruses, worms, ransomware, spyware.

Q59 Concept Language Translator

Which of the following generates intermediate code?

A. Assembler

B. Linker

C. Compiler

D. Loader

Ans: C A compiler generates intermediate code (e.g., bytecode in Java) as part of compilation.

Q60 Statement Algorithm

A flowchart MUST always be drawn before writing an algorithm. This statement is:

A. Always true

B. False — an algorithm can be expressed as pseudo-code or flowchart, either is valid

C. True for competitive exams only

D. True only in structured programming

Ans: B False — an algorithm can be expressed as a flowchart OR pseudo-code. Neither is mandatory first.

Q96 Application MS Office

Which MS Excel function returns the current date and time?

A. =TODAY()

B. =NOW()

C. =DATE()

D. =TIME()

Ans: B =NOW() returns the current date and time. =TODAY() returns only the current date.

Q97 Concept Software Types

An ERP (Enterprise Resource Planning) system is an example of:

A. General-purpose software

B. Specific-purpose application software

C. System software

D. Firmware

Ans: B ERP systems (SAP, Oracle) are specific-purpose application software designed for enterprise business processes.

Q98 Concept Algorithm

Which of the following traversal methods for a binary tree visits nodes in Left-Root-Right order?

A. Pre-order

B. In-order

C. Post-order

D. Level-order

Ans: B In-order traversal visits Left, then Root, then Right (LRootR). For BST, this gives sorted output.

Q99 Concept High-Level Language

Which feature of OOP restricts direct access to object components?

A. Inheritance

B. Polymorphism

C. Encapsulation

D. Abstraction

Ans: C Encapsulation bundles data and methods together and restricts direct access using access modifiers (private, public).

Q100 Application MS Office

Which MS Office feature automatically corrects common spelling errors as you type?

A. Spell Check

B. AutoCorrect

C. Grammar Check

D. AutoComplete

Ans: B AutoCorrect automatically fixes common typos and spelling errors as you type in MS Office.

Q101 Concept Flowchart

Which of the following is an advantage of using a flowchart?

A. It directly executes as code

B. It provides a clear visual representation of program logic

C. It replaces the need for an algorithm

D. It works only for simple programs

Ans: B Flowcharts provide a clear visual representation of logic, aiding communication and debugging.

Q102 Concept Algorithm

A stack data structure follows which principle?

A. FIFO — First In First Out

B. LIFO — Last In First Out

C. Random Access

D. Priority-based access

Ans: B A stack follows LIFO (Last In First Out). Push adds to top; Pop removes from top.

Q103 Concept System Software

Multitasking in an operating system refers to:

A. Running multiple CPUs

B. Running multiple processes seemingly simultaneously on a single CPU

C. Using multiple RAM modules

D. Running software on multiple computers

Ans: B Multitasking allows the OS to run multiple processes concurrently by rapidly switching CPU time between them.

Q104 | Concept | Language Translator

Which programming language is typically used for scripting in web browsers?

A. Java

B. C++

C. JavaScript

D. Python

Ans: C JavaScript is the primary scripting language for web browsers, enabling dynamic web pages.

Q105 | Application | MS Office

In MS Excel, the IF function syntax is:

A. =IF(condition, value_if_true, value_if_false)

B. =IF(value_if_true, condition, value_if_false)

C. =IF(value_if_false, value_if_true, condition)

D. =IF(condition; value1; value2)

Ans: A Correct syntax: =IF(logical_test, value_if_true, value_if_false). Example: =IF(A1>10,"Pass","Fail").

Q119 | Concept | High-Level Language

Which language is primarily used for statistical computing and data analysis?

A. Java

B. R

C. C++

D. Assembly

Ans: B R is a programming language designed for statistical computing and data analysis, widely used in data science.

Q120 | Concept | Software Types

An antivirus program is classified as:

A. Application software

B. Utility software (a type of system software)

C. Firmware

D. Middleware

Ans: B Antivirus is utility software — a subcategory of system software that maintains and protects the computer.

Q121 | Application | Algorithm

For a sorted array of 1024 elements, binary search requires at most how many comparisons?

A. 10

B. 11

C. 512

D. 1024

Ans: A log2(1024) = 10 comparisons maximum. Binary search halves the search space each step.

Q122 | Concept | Stored Program Concept

The Arithmetic Logic Unit (ALU) is responsible for:

A. Storing programs in memory

B. Performing arithmetic and logical operations

C. Controlling the sequence of instructions

D. Managing input/output

Ans: B The ALU performs all arithmetic (add, subtract, multiply) and logical (AND, OR, NOT) operations.

Q123 | Concept | MS Office

In MS Word, 'Track Changes' feature is used to:

A. Change font automatically

B. Record edits made to a document for review

C. Track cursor movements

D. Print documents

Ans: B Track Changes records insertions, deletions, and formatting changes made to a document, useful for collaborative editing.

Q141 | Concept | System Software

Spooling (Simultaneous Peripheral Operations On-Line) is used to:

A. Speed up the CPU

B. Manage slow I/O devices by queuing jobs (e.g., print queue)

C. Increase RAM size

D. Compress files on disk

Ans: B Spooling queues I/O tasks (especially printing) so that the CPU doesn't have to wait for slow peripheral devices. Print spooling is the most common example.

Q142 | Concept | Language Translator

Which of the following is NOT a phase of a compiler?

A. Lexical Analysis

B. Syntax Analysis

C. Execution

D. Code Generation

Ans: C

Execution is not a compiler phase. Compiler phases: Lexical Analysis → Syntax Analysis → Semantic Analysis → Intermediate Code Generation → Optimization → Code Generation.

Q143 | Application | Algorithm

What is the worst-case time complexity of Quick Sort?

A. O(n log n)

B. O(n)

C. O(n^2)

D. O(log n)

Ans: C

Quick Sort's worst case is O(n^2), which occurs when the pivot is always the smallest or largest element (e.g., already sorted array with naive pivot selection).

Q144 | Assertion-Reason | Firmware

Assertion (A): Firmware updates can 'brick' a device if interrupted.
Reason (R): Firmware is written to non-volatile flash memory; a failed write can leave the device unbootable.

A. Both A and R are true, R correctly explains A

B. A is true but R is false

C. Both are false

D. A is false, R is true

Ans: A

Both are true and R explains A. If firmware flashing is interrupted (power failure, crash), the flash memory may be partially written, leaving no valid boot firmware — 'bricking' the device.

Q145 | Concept | High-Level Language

Which of the following correctly describes 'abstraction' in OOP?

A. Hiding the implementation and showing only the functionality to the user

B. Allowing one class to acquire properties of another

C. A single method having multiple forms

D. Binding data and functions together

Ans: A Abstraction hides complex implementation details and exposes only what is necessary. It is implemented using abstract classes and interfaces.

Q146 | Application | MS Office

In MS Excel, which function returns the largest value in a range?

A. =LARGE()

B. =MAX()

C. =HIGH()

D. =TOP()

Ans: B =MAX(range) returns the largest numeric value in a specified range. =LARGE(range, k) returns the k-th largest value.

Q147 | Statement | Algorithm

Statement 1: A queue follows FIFO (First In First Out) discipline.
Statement 2: A stack follows FIFO discipline.
Which is/are correct?

A. Only Statement 1

B. Only Statement 2

C. Both are correct

D. Both are incorrect

Ans: A Only Statement 1 is correct. A queue follows FIFO. A stack follows LIFO (Last In First Out), not FIFO.

Q148 | Concept | Software Types

Which of the following is a characteristic of proprietary software?

A. Source code is freely available

B. Can be freely distributed and modified

C. Source code is kept secret; users must pay a license fee

D. Developed by volunteer communities

Ans: C Proprietary (closed-source) software keeps its source code secret and typically requires payment of a license fee. Examples: Microsoft Windows, Adobe Photoshop.

Q149 | Application | MS Office

Which of the following is NOT a valid chart type in MS Excel?

A. Bar chart

B. Pie chart

C. Scatter chart

D. Paragraph chart

Ans: D 'Paragraph chart' is not a chart type. MS Excel supports Bar, Pie, Scatter, Line, Column, Area, Doughnut, Radar, and other standard chart types.

Q150 | Concept | Stored Program Concept

The Memory Address Register (MAR) in a CPU holds:

A. The result of the last arithmetic operation

B. The address of the memory location being accessed

C. The current instruction being executed

D. The next instruction to be fetched

Ans: B MAR holds the address of the memory location from which data is to be read or to which data is to be written. It interfaces between CPU and memory bus.

Advanced Level — 52 Questions

Q61 Concept — Stored Program Concept

In von Neumann architecture, the bottleneck between CPU and memory is called:

A. Cache miss

B. Von Neumann bottleneck

C. Pipeline stall

D. Memory leak

Ans: B The von Neumann bottleneck refers to limited data transfer between CPU and memory sharing the same bus.

Q62 Concept — Language Translator

A JIT (Just-In-Time) compiler:

A. Compiles code before execution

B. Compiles code at runtime, combining interpreter and compiler benefits

C. Only interprets code

D. Is used for assembly language

Ans: B JIT compilation occurs at runtime — combines flexibility of interpretation with speed of compilation (Java JVM, .NET CLR).

Q63 Application — Algorithm

Which algorithm design paradigm does merge sort use?

A. Greedy

B. Dynamic Programming

C. Divide and Conquer

D. Backtracking

Ans: C Merge Sort uses Divide and Conquer — recursively divides, sorts sub-arrays, then merges. O(n log n).

Q64 Statement — High-Level Language

Statement 1: Python is an interpreted language.
Statement 2: Java source code is compiled to bytecode, then interpreted by JVM.
Which is/are correct?

A. Only Statement 1

B. Only Statement 2

C. Both are correct

D. Both are incorrect

Ans: C Both correct. Python is interpreted. Java is compiled to platform-independent bytecode, then JVM interprets/JIT-compiles.

Q65 Assertion-Reason — High-Level Language

Assertion (A): High-level languages are machine-independent.
Reason (R): HLL programs need a translator to convert to machine code.

A. Both A and R are true, R correctly explains A

B. Both A and R are true, R does not explain A

C. A is true, R is false

D. A is false, R is true

Ans: A HLL programs run on any machine with a suitable translator (A true). Because they need translation, they are machine-independent (R explains A).

Q66 Concept — Firmware

Device drivers differ from firmware because:

A. Firmware is software, drivers are hardware

B. Drivers run in OS context and can be updated via OS; firmware is embedded in hardware

C. Firmware is in RAM, drivers are in ROM

D. Both are identical

Ans: B Drivers are OS-level software, installable/updatable via OS. Firmware is embedded in hardware (ROM/Flash).

Q67 Application — Stored Program Concept

In the fetch-decode-execute cycle, which register holds the address of the next instruction?

A. Accumulator

B. Program Counter (PC)

C. Memory Address Register (MAR)

D. Instruction Register (IR)

Ans: B The Program Counter (PC) holds the memory address of the next instruction to fetch, then increments.

Q68 Concept — Language Generations

5th generation languages (5GL) are primarily based on:

A. Object-oriented programming

B. Logic and constraints (AI-based)

C. Procedural programming

D. Assembly

Ans: B 5GLs (Prolog, LISP for AI) are based on solving constraints and logic — programmer defines WHAT, not HOW.

Q69 Application — Algorithm

An algorithm has recurrence T(n) = 2T(n/2) + n. Its time complexity is:

A. O(n)

B. O(n log n)

C. O(n^2)

D. O(log n)

Ans: B By Master Theorem (case 2): T(n) = O(n log n). This is Merge Sort's complexity.

Q70 | Match | Software

Match the software category with the example:
P. System Software Q. Application Software R. Firmware S. Utility Software
1. Norton Antivirus 2. BIOS 3. Windows OS 4. MS Excel

A. P-3, Q-4, R-2, S-1

B. P-4, Q-3, R-1, S-2

C. P-2, Q-1, R-3, S-4

D. P-3, Q-2, R-4, S-1

Ans: A Windows OS=System; MS Excel=Application; BIOS=Firmware; Norton Antivirus=Utility Software.

Q71 | Concept | Stored Program Concept

Harvard architecture differs from von Neumann architecture in that:

A. It uses no memory

B. It has separate memory and buses for instructions and data

C. It uses only one register

D. It does not use a CPU

Ans: B Harvard architecture uses physically separate storage for instructions and data, avoiding von Neumann bottleneck.

Q72 | Application | Software

Which of the following testing types checks if different modules work correctly together?

A. Unit Testing

B. Integration Testing

C. System Testing

D. Acceptance Testing

Ans: B Integration Testing verifies that individual modules work correctly when combined.

Q73 Concept — System Software

The role of a linker in software development is to:

A. Translate source code to machine code

B. Combine object files and libraries into a single executable

C. Load programs into memory

D. Debug the program

Ans: B A linker combines multiple object files and resolves library references, producing a single executable file.

Q74 Concept — System Software

A loader is responsible for:

A. Translating source code

B. Linking object modules

C. Loading the executable program into RAM for execution

D. Compiling the program

Ans: C The loader is part of the OS that loads an executable into RAM so the CPU can execute it.

Q75 Concept — Algorithm

Which of the following problems is NP-Complete?

A. Sorting

B. Binary Search

C. Travelling Salesman Problem (decision version)

D. Matrix multiplication

Ans: C TSP decision version is NP-Complete — no known polynomial-time algorithm. Sorting/searching are polynomial.

Q76 Concept — Language Translator

The symbol table in a compiler is used to:

A. Store machine code

B. Store information about identifiers (variables, functions, types)

C. Generate object code

D. Handle runtime errors

Ans: B The symbol table stores identifier info — name, type, scope, memory address — used throughout compilation.

Q77 Application Software

In SDLC, which phase comes immediately after System Analysis?

A. Coding

B. Testing

C. System Design

D. Implementation

Ans: C SDLC: Planning -> Analysis -> Design -> Implementation -> Testing -> Deployment -> Maintenance.

Q78 Application Algorithm

Which sorting algorithm has the best average-case time complexity?

A. Bubble Sort - O(n^2)

B. Selection Sort - O(n^2)

C. Quick Sort - O(n log n)

D. Insertion Sort - O(n^2)

Ans: C Quick Sort has average O(n log n) — fastest in practice. Bubble, Selection, Insertion are O(n^2) on average.

Q79 Concept Language Translator

Semantic analysis in a compiler checks:

A. Spelling of keywords

B. Grammar rules

C. Meaning and type consistency (e.g., type checking)

D. Machine code generation

Ans: C Semantic analysis verifies type checking, scope resolution, and that operations are valid.

Q80 Concept System Software

Virtual memory allows a computer to:

A. Increase ROM size

B. Use secondary storage as an extension of RAM

C. Speed up the CPU

D. Replace cache memory

Ans: B Virtual memory uses hard disk space as an extension of RAM (via paging/swapping), managed by the OS.

Q81 | Statement | Stored Program Concept

In the von Neumann model, the Control Unit (CU):
(I) Controls the sequence of operations.
(II) Performs arithmetic calculations.

A. Only I is correct

B. Only II is correct

C. Both are correct

D. Neither is correct

Ans: A CU controls/coordinates the sequence (fetch, decode, execute). Arithmetic is done by ALU, not CU.

Q82 | Application | Algorithm

Dynamic programming differs from divide-and-conquer in that:

A. DP cannot solve optimization problems

B. DP stores and reuses solutions to overlapping subproblems (memoization)

C. D&C; always has overlapping subproblems

D. DP never divides problems

Ans: B DP stores solutions to subproblems (memoization/tabulation) to avoid redundant computation.

Q83 | Concept | Language Translator

A cross-compiler is one that:

A. Runs on a computer and generates code for the same machine

B. Runs on one platform and generates code for a different platform

C. Compiles two languages simultaneously

D. Is faster than a normal compiler

Ans: B A cross-compiler runs on a host platform but generates executable code for a different target platform.

Q84 Assertion-Reason High-Level Language

Assertion (A): Python is platform-independent.
Reason (R): Python code is interpreted by a Python interpreter available for multiple OS platforms.

A. Both A and R are true, R correctly explains A

B. A is true but R is false

C. Both are false

D. A is false, R is true

Ans: A Python is platform-independent (A); because the interpreter exists for all platforms, the same code runs anywhere (R explains A).

Q85 Concept Algorithm

Which sorting algorithm has worst-case O(n^2) but is adaptive (best case O(n))?

A. Merge Sort

B. Quick Sort

C. Insertion Sort

D. Heap Sort

Ans: C Insertion Sort is adaptive — O(n) when already sorted, O(n^2) worst case. Ideal for small/nearly sorted data.

Q106 Concept Algorithm

Which data structure is best suited for implementing a priority queue?

A. Array

B. Linked List

C. Heap

D. Stack

Ans: C A Heap (min-heap or max-heap) is the most efficient structure for a priority queue: O(log n) insert/delete.

Q107 Application Software

In Agile methodology, a 'Sprint' refers to:

A. A software testing phase

B. A fixed time-boxed development iteration (typically 1-4 weeks)

C. A type of algorithm

D. A deployment pipeline

Ans: B In Agile/Scrum, a Sprint is a fixed-length iteration (usually 1-4 weeks) during which a product increment is developed.

Q108 Concept Language Translator

Lexical analysis in a compiler converts source code into:

A. Parse trees

B. Tokens (lexemes)

C. Object code

D. Symbol tables

Ans: B Lexical analysis (scanning) breaks the source code into tokens — identifiers, keywords, operators, literals.

Q109 Application Algorithm

The time complexity of Heap Sort is:

A. O(n)

B. O(n log n)

C. O(n^2)

D. O(log n)

Ans: B Heap Sort has O(n log n) time complexity in all cases (best, average, worst) — unlike Quick Sort's O(n^2) worst case.

Q110 Concept System Software

Thrashing in an operating system occurs when:

A. CPU usage is very high and productive

B. The OS spends more time paging than executing processes

C. Multiple CPUs are used simultaneously

D. The system runs out of ROM

Ans: B Thrashing occurs when excessive page faults cause the OS to spend more time swapping pages than executing useful work.

Q111 Concept High-Level Language

Which of the following best describes 'polymorphism' in OOP?

A. One class inheriting from another

B. Hiding implementation details

C. One interface having multiple implementations

D. Grouping data and functions together

Ans: C Polymorphism means one interface/method having multiple forms — method overloading (compile-time) and overriding (runtime).

Q112 Application Software

The Waterfall model is characterized by:

A. Iterative development cycles

B. Strictly sequential phases with no going back

C. Continuous delivery

D. Rapid prototyping

Ans: B The Waterfall model has sequential phases (Requirements -> Design -> Implementation -> Testing -> Maintenance) with no backtracking.

Q113 Concept Algorithm

Dijkstra's algorithm is used to find:

A. Maximum flow in a network

B. Shortest path from a source to all other vertices in a weighted graph

C. Minimum spanning tree

D. Topological ordering

Ans: B Dijkstra's algorithm finds the shortest path from a single source to all vertices in a graph with non-negative weights.

Q114 Concept — System Software

Which scheduling algorithm is used by most modern OS for interactive processes?

A. First Come First Serve (FCFS)

B. Shortest Job First (SJF)

C. Round Robin

D. Priority Scheduling (non-preemptive)

Ans: C Round Robin scheduling gives each process a fixed time quantum in cyclic order, suitable for interactive time-sharing systems.

Q115 Concept — Language Translator

An optimizing compiler differs from a regular compiler in that it:

A. Translates faster

B. Improves code efficiency (speed/memory) without changing its behavior

C. Supports more programming languages

D. Generates human-readable code

Ans: B An optimizing compiler applies transformations (dead code elimination, loop unrolling, inlining) to generate faster/smaller code.

Q124 Application — Algorithm

What is the space complexity of the recursive implementation of Fibonacci using memoization?

A. O(1)

B. O(n)

C. O(n^2)

D. O(2^n)

Ans: B With memoization, both time and space complexity become O(n) — storing n computed values in a memo table.

Q125 Concept Software

In software testing, 'White Box Testing' means:

A. Testing without any knowledge of internal code

B. Testing with full knowledge of internal code structure

C. Testing only the user interface

D. Testing by end users

Ans: B White Box (glass box/structural) testing uses knowledge of internal code to design test cases covering code paths.

Q126 Concept High-Level Language

Which of the following is a functional programming language?

A. C

B. Java

C. Haskell

D. Assembly

Ans: C Haskell is a purely functional programming language. C and Java are imperative/OOP; Assembly is low-level.

Q127 Application MS Office

In MS Excel, what does the $ symbol do in a cell reference like A1?

A. Represents currency

B. Makes the reference absolute (does not change when copied)

C. Locks the worksheet

D. Marks a formula error

Ans: B The $ sign creates an absolute reference — A1 does not change row or column when the formula is copied.

Q128 Concept System Software

Deadlock in operating systems occurs when:

A. A process uses too much memory

B. Two or more processes wait indefinitely for resources held by each other

C. The CPU is overloaded

D. A program crashes

Ans: B Deadlock occurs when processes form a circular wait — each holds a resource the other needs, and none can proceed.

Q129 Concept: Algorithm

Prim's and Kruskal's algorithms are used to find:

A. Shortest path

B. Maximum flow

C. Minimum Spanning Tree (MST)

D. Topological sort

Ans: C Both Prim's and Kruskal's algorithms find the Minimum Spanning Tree of a weighted undirected graph.

Q130 Concept: Software

Software that is distributed under a license requiring any derivative works to be released under the same license is called:

A. MIT license

B. BSD license

C. Copyleft (GPL) license

D. Proprietary license

Ans: C Copyleft (e.g., GNU GPL) requires derivative works to carry the same license — 'share-alike' clause. MIT/BSD are permissive.

Q151 Concept: Algorithm

In graph theory, topological sorting can be applied to:

A. Any undirected graph

B. Only trees

C. Directed Acyclic Graphs (DAGs) only

D. Only weighted graphs

Ans: C Topological sort is only valid for Directed Acyclic Graphs (DAGs). It produces a linear ordering of vertices such that for every edge u→v, u comes before v.

Q152 | Application | Software

In software engineering, 'refactoring' means:

A. Rewriting software from scratch

B. Restructuring existing code without changing its external behavior to improve readability/maintainability

C. Adding new features to software

D. Testing software after deployment

Ans: B Refactoring improves the internal structure of code (readability, simplicity, performance) without altering its observable behavior. It is not about adding features.

Q153 | Concept | Language Translator

In compiler design, 'peephole optimization' operates on:

A. The entire program at once

B. A small window of consecutive instructions to replace them with more efficient equivalents

C. Only loop structures

D. The symbol table entries

Ans: B Peephole optimization examines a small 'window' (peephole) of generated code at a time and replaces instruction sequences with shorter or faster equivalents (e.g., removing redundant loads).

Q154 | Concept | System Software

The 'Banker's Algorithm' in operating systems is used to:

A. Manage file allocation

B. Avoid deadlock by ensuring the system stays in a safe state before granting resources

C. Schedule CPU processes

D. Manage virtual memory paging

Ans: B The Banker's Algorithm (Dijkstra) is a deadlock avoidance algorithm. Before allocating resources, it checks whether the resulting state is 'safe' (all processes can eventually complete).

Q155 Application — Algorithm

Which data structure is most efficient for implementing an LRU (Least Recently Used) cache?

A. Array

B. Stack

C. HashMap + Doubly Linked List

D. Binary Search Tree

Ans: C An LRU cache is optimally implemented using a HashMap (for O(1) lookups) combined with a Doubly Linked List (for O(1) insertions/deletions to track recency).

Q156 Concept — High-Level Language

In Python, what is a 'decorator'?

A. A type of loop

B. A class that inherits from another

C. A function that takes another function as argument and extends its behavior without modifying it

D. A built-in sorting function

Ans: C A Python decorator is a higher-order function that wraps another function to add behavior (logging, authentication, timing) without modifying the original function's code.

Q157 Assertion-Reason — System Software

Assertion (A): In a multithreaded program, a mutex (mutual exclusion lock) prevents race conditions.
Reason (R): A mutex ensures only one thread accesses a shared resource at a time.

A. Both A and R are true, R correctly explains A

B. A is true but R is false

C. Both are false

D. A is false, R is true

Ans: A Both true. A mutex prevents race conditions (A) by allowing only one thread to hold the lock at a time, serializing access to shared resources (R explains A).

Q158 Application — Algorithm

Floyd-Warshall algorithm is used to find:

A. Single-source shortest path

B. Minimum spanning tree

C. All-pairs shortest paths in a weighted graph

D. Maximum flow in a network

Ans: C Floyd-Warshall finds the shortest paths between ALL pairs of vertices in O(V^3) time. It works with negative edges (but not negative cycles). Dijkstra is single-source only.

Q159 Concept — Software

Which software design pattern provides a simplified interface to a complex subsystem?

A. Observer

B. Singleton

C. Facade

D. Factory

Ans: C The Facade pattern provides a simplified, unified interface to a set of interfaces in a complex subsystem, making it easier to use without exposing internal complexity.

Q160 Statement — Stored Program Concept

Consider the statements about RISC vs CISC architectures:
(I) RISC has a small set of simple, fixed-length instructions.
(II) CISC has complex instructions that can perform multi-step operations.
Which is/are correct?

A. Only I

B. Only II

C. Both I and II

D. Neither

Ans: C Both correct. RISC (Reduced Instruction Set Computer) uses simple, uniform instructions for fast pipelining. CISC (Complex Instruction Set Computer) uses variable-length, multi-step instructions like x86.

Answer Key — All 130 Questions

Q#	1	2	3	4	5	6	7	8	9	10
1–10	B	B	C	B	B	B	C	C	C	C
11–20	B	C	C	D	B	B	C	D	C	C
21–30	B	C	B	A	B	B	B	B	B	B
31–40	B	B	B	B	C	C	C	C	B	B
41–50	B	B	B	A	B	C	B	B	B	B
51–60	B	C	C	B	C	B	C	B	C	B
61–70	B	B	C	C	A	B	B	B	B	A
71–80	B	B	B	C	C	B	C	C	C	B
81–90	A	B	B	A	C	B	B	B	B	B
91–100	B	C	D	B	C	B	B	B	C	B
101–110	B	B	B	C	A	C	B	B	B	B
111–120	C	B	B	C	B	D	B	B	B	B
121–130	A	B	B	B	B	C	B	B	C	C

Color Code: Green shading = Basic Amber shading = Intermediate Red shading = Advanced